Reg. No.														

B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Fifth Semester

21CSC302J - COMPUTER NETWORKS

(For the candidates admitted from the academic year2021-2022 to 2023-2024)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer **ALL** Questions

Q.	Question	Marks	BL	CO	PO
1.	Suppose 6 devices are connected with each other in a mesh topology network, then what is the total number of dedicated links required to connect them? A) 15 B) 14 C) 9 D) 8	1	3	1	1
2.	Communication between the traditional monitor and the computer involves __________ mode of transmission. A) Full-duplex B) Half-duplex C) Simplex D) Automatic	1	1	1	1
3.	Match the following 1. Router i) Data Link Layer 2. Hub ii) Network Layer 3. Bridge iii) All Layers 4. Gateway iv) Physical Layer A) 1 – iv, 2 – iii, 3 – i, 4 -ii B) 1 – ii, 2 – iv, 3 –i, 4 - iii C) 1 - iii, 2 – iv, 3 – ii, 4 – i D) 1 – i, 2 – iii, 3 – ii, 4 – iv	1	2	1	1
4.	Layer 2 groups together the raw data (1's and 0's) into chunks, known as ___________. A) Packets B) Datagrams C) Segments D) Frames	1	1	1	1
5.	Find the error, if any, in the following IPv4 addresses. i. 11.56.045.78 ii. 221.34.7.8.20 iii. 75.45.256.14 iv. 10.23.14.01 A) Only options i ,ii and iii are having errors B) Only options ii , iii, and iv are having errors C) All the 4 options are having errors D) All the 4 options are correct	1	3	2	1
6.	A router receives a packet with the destination address 132.168.16.4/20. Find the broadcast address for the network. A) 132.168.31.255 B) 132.168.15.255 C) 132.168.16.255 D) 132.168.255.255	1	1	2	4

7. __________ field in the IP header is used to specify how long a datagram can reside in a network before it reaches the destination. 1 1 2 1

A) Identifications
B) Options
C) Type of Service (TOS)
D) Time To Live (TTL)

8. _________ is a 2 port device used for interconnecting two LANs working on the same protocol. 1 2 2 1

A) Switch
B) Bridge
C) Router
D) Modem

9. How does the Distance Vector Routing algorithm prevent routing loops? 1 1 3 4

A) By sending periodic Hello messages
B) By implementing the Split-Horizon rule
C) By using the Link-State Advertisement.
D) By using the Spanning Tree Protocol.

10. What algorithm is commonly used by Link State Routing protocols to compute the shortest path? 1 1 3 1

A) Floyd-Warshall algorithm
B) Bellman-Ford algorithm
C) A* algorithm
D) Dijkstra's algorithm

11. Which protocol is an example of a Path Vector Routing protocol? 1 1 3 1

A) BGP
B) RIP v1
C) OSPF
D) RIP v2

12. Which of the following attributes is used by BGP to determine the best path when multiple paths are available? 1 2 3 1

A) Bandwidth
B) Hop count
C) AS_Path
D) Delay

13. In CSMA/CD, what does a device do when it detects a collision on the network? 1 2 4 1

A) It stops transmitting and sends a jamming signal.
B) It immediately retransmits the data.
C) It continues transmitting but at a lower power level.
D) It waits for a fixed time and then retransmits

14. What is the primary function of the Ethernet frame's MAC (Media Access Control) address? 1 2 4 4

A) To segment the frame data into smaller packets.
B) To encrypt the frame data for secure transmission.
C) To identify the source and destination of the frame within a local network.
D) To route the frame to the correct IP address.

15. What is the Hamming distance between the binary strings `101010` and `100100`? 1 2 4 4

A) 2
B) 3
C) 4
D) 5

16. What is the primary difference between Stop-and-Wait ARQ and Sliding Window ARQ? 1 2 4 4

A) Stop-and-Wait ARQ allows multiple packets to be sent before requiring an acknowledgment, while Sliding Window ARQ only sends one packet at a time.
B) Stop-and-Wait ARQ uses a sliding window mechanism, while Sliding Window ARQ uses a stop-and-wait mechanism
C) Sliding Window ARQ is used in analog communication, while Stop-and-Wait ARQ is used in digital communication.
D) Sliding Window ARQ allows multiple packets to be sent before requiring an acknowledgment, while Stop-and-Wait ARQ only sends one packet at a time.

17.	Which of the following protocols is commonly used with UDP for real-time applications? A) Hyper Text Transfer Protocol (HTTP) B) File Transfer Protocol (FTP) C) Real-time Transport Protocol (RTP) D) Simple Mail Transfer Protocol (SMTP)	1	2	5	1
18.	What mechanism does TCP use to ensure reliable data transmission? A) Time-to-live (TTL) B) Retransmission of lost packets C) Packet filtering D) Data encryption	1	1	5	1
19.	Which HTTP status code indicates that a request was successful and the server returned the requested resource? A) 200 OK B) 404 Not Found C) 500 Internal Server Error D) 301 Moved Permanently	1	2	5	1
20.	Which protocol is used by Telnet for communication? A) ICMP B) UDP C) IP D) TCP	1	1	5	1

PART - B (5 x 8 = 40 Marks)

Answer **ALL** Questions

		Marks	BL	CO	PO
21 a.	Distinguish between Circuit Switching and Packet Switching techniques.	8	3	1	1
	(OR)				
b.	Explain about different network devices used in the computer network.	8	3	1	1
22 a.	Each of the following addresses belongs to a block. Find the first and the last address in each block. (i). 14.12.72.8/24 (ii). 200.107.16.17/18	8	4	2	4
	(OR)				
b.	Discuss the Network Address Translation (NAT) in terms of (i) using a pool of IP addresses (ii) Using Both IP Addresses and Port Addresses	8	4	2	1
23 a.	Explain what type of OSPF link state is advertised in each of the following cases: (i) A router needs to advertise the existence of another router at the end of a point-to-point link (ii) A designated router advertises a network as a node	8	3	3	4
	(OR)				
b.	In a small network, three routers R1, R2, and R3 are connected in a triangular topology. Each router uses the distance-vector routing algorithm, like RIP (Routing Information Protocol), to share its routing table with its directly connected neighbours every 30 seconds. One day, the link between R1 and R2 fails. However, due to the nature of the distance-vector algorithm, R1 and R2 are not immediately aware of the link failure. R3 continues to send routing updates to R1 and R2 advertising a route to each other through itself. (i) How might a routing loop form in this network following the link failure between **R1** and **R2**? (ii) Describe the mechanisms employed by the distance-vector routing protocol to prevent the formation of this routing loop.	8	3	3	4
24 a.	Explain the process of collision detection and resolution in CSMA/CD.	8	3	4	1
	(OR)				

	Question	Marks	BL	CO	PO
	b. Describe how the Go-Back-N and Selective Repeat mechanisms work in the context of the Sliding Window ARQ protocol. What are the trade-offs between these two approaches?	8	3	4	1
25	a. Compare and contrast TCP and UDP.	8	2	5	1
	(OR)				
	b. What are the different types of DNS records? Explain their roles in DNS resolution.	8	2	5	1

PART - C (1 x 15 = 15 Marks)

Answer **ANY ONE** Question

	Question	Marks	BL	CO	PO
26.	(i) Given the IP address block 192.168.1.0/24, how would you subnet it to accommodate the following requirements: (a) 2 subnets with at least 30 hosts each (b) 3 subnets with at least 10 hosts each (ii) If you need to create 5 subnets from the 10.0.0.0/24 block, with each subnet having at least 20 hosts, what will be the subnet mask and address ranges?	15	4	2	4
27.	Consider a network using the Distance Vector routing protocol (e.g., RIP). The network initially consists of three routers, A, B, and C, with the following connections: • A <-> B (cost 1) • B <-> C (cost 2) • A <-> C (cost 5) Router A wants to add a new connection to Router D with a cost of 3. How will the network update its routing tables, how will the routes to Router D be calculated by each router, and state the Distance Vector Routing Algorithm	15	4	5	4

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